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4 / 4 submitted

Level 1 — 100 marks/question

L1-Q1. hexadecimal (E)

C

CODE

```
#include<stdio.h>

int main(){
    char s[100];
    long long dec = 0;
    int i = 0;

    scanf("%99s",s);

    while(s[i]){
        dec *=16;
        char c = s[i++];
        if('0' ≤ c && c ≤ '9' )
            dec += c - '0';
        else if('a' ≤ c && c ≤ 'f')
            dec += c-'a'+10;
        else if('A' ≤ c && c ≤ 'F')
            dec +=c -'A'+10;
    }
    printf("Decimal equivalent: %lld\n",dec);
    return 0;
}
```

INPUT

FF

OUTPUT

Decimal equivalent: 255

L1-Q2. decimal (E)

C

CODE

```
#include<stdio.h>

int main(){
    int n,binary[32],i=0;
    scanf("%d",&n);

    while(n>0){
        binary[i++] = n%2;
        n /= 2;
    }
    printf("Binary equivalent: ");
    for(int j=i-1;j≥0;j--){
        printf("%d",binary[j]);
    }
    printf("\n");
    return 0;
}
```

INPUT

2

OUTPUT

Binary equivalent: 10

L1-Q3. Armstrong (E)

C

CODE

```
#include<stdio.h>
#include<math.h>

int main(){
    int n,temp,digits=0,sum=0,rem;

    scanf("%d",&n);

    temp=n;
    while(temp != 0){
        digits ++;
        temp /= 10;
    }
    temp=n;
    while(temp !=0){
        rem = temp%10;
        sum += pow(rem,digits);
        temp /= 10;
    }
    if(sum == n){
        printf("%d is an Armstrong number",n);}
    else{
        printf("%d is not an Armstrong number",n);
    }
    return 0;
}
```

INPUT

123

OUTPUT

123 is not an Armstrong number

L1-Q4. Hollow Square (E)

C

CODE

```
#include<stdio.h>
int main(){
    int n=5,i,j;

    for(i=0;i<n;i++){
        for(j=0;j<n;j++){
            if(i==0||i==n-1||j==0||j==n-1){
                printf("*");
            }
            else{
                printf(" ");
            }
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT

```
*****
*  *
*  *
*  *
*****
```